

# UniShield — IP Addressing Scheme

Complete IP addressing for all network interfaces (Routers, FortiGate, Wazuh, Switches)

| Device     | Interface     | IP Address      | Subnet Mask     | Connected To      | Purpose                          |
|------------|---------------|-----------------|-----------------|-------------------|----------------------------------|
| R1         | Fa0/0 (trunk) | —               | —               | FACULTY(1) switch | VLAN trunk — no IP on physical   |
| R1         | Fa0/0.10      | 192.168.10.1    | 255.255.255.0   | VLAN 10 hosts     | Gateway for Faculty VLAN 10      |
| R1         | Fa0/0.20      | 192.168.20.1    | 255.255.255.0   | VLAN 20 hosts     | Gateway for Faculty VLAN 20      |
| R1         | Fa0/0.30      | 192.168.30.1    | 255.255.255.0   | VLAN 30 hosts     | Gateway for Faculty VLAN 30      |
| R1         | Fa1/0         | 20.0.0.1        | 255.255.255.252 | R2 Fa1/0          | WAN Link 1 — /30                 |
| R1         | Fa2/0         | 10.0.0.1        | 255.255.255.252 | R2 Fa2/0          | WAN Link 2 — /30                 |
| R1         | Fa3/0         | 11.0.0.2        | 255.255.255.0   | FortiGate port2   | Uplink to NGFW                   |
| R1         | Fa4/0         | 10.10.10.15     | 255.255.255.0   | Cloud0 / Wazuh    | SIEM management bridge           |
| R1         | Fa5/0         | 30.0.0.30       | 255.0.0.0       | VoIP network      | VoIP gateway                     |
| R2         | Fa0/0 (trunk) | —               | —               | FACULTY_2 switch  | VLAN trunk — no IP on physical   |
| R2         | Fa0/0.40      | 192.168.40.1    | 255.255.255.0   | VLAN 40 hosts     | Gateway for Students VLAN        |
| R2         | Fa0/0.50      | 192.168.50.1    | 255.255.255.0   | VLAN 50 hosts     | Gateway for Management VLAN      |
| R2         | Fa0/0.60      | 192.168.60.1    | 255.255.255.0   | VLAN 60 hosts     | Gateway for Staff VLAN           |
| R2         | Fa1/0         | 20.0.0.2        | 255.255.255.252 | R1 Fa1/0          | WAN Link 1 — /30                 |
| R2         | Fa2/0         | 10.0.0.2        | 255.255.255.252 | R1 Fa2/0          | WAN Link 2 — /30                 |
| R2         | Fa4/0         | 12.0.0.2        | 255.255.255.0   | FortiGate port3   | Uplink to NGFW                   |
| FortiGate  | port1         | 192.168.183.130 | 255.255.255.0   | VMware NAT        | WAN / Internet                   |
| FortiGate  | port2         | 11.0.0.1        | 255.255.255.0   | R1 Fa3/0          | Internal side 1 — Faculty        |
| FortiGate  | port3         | 12.0.0.1        | 255.255.255.0   | R2 Fa4/0          | Internal side 2 — Admin/Students |
| Wazuh SIEM | eth0          | 10.10.10.100    | 255.255.255.0   | Cloud0 bridge     | SIEM UDP 514 listener            |
| FACULTY(1) | Vlan10 SVI    | 192.168.10.2    | 255.255.255.0   | Management        | Switch management IP             |
| FACULTY_2  | Vlan40 SVI    | 192.168.40.2    | 255.255.255.0   | Management        | Switch management IP             |